



Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

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Laboratory Manual

Academic Year	:	2026-27	Semester	:	3rd
Course code	:	ITUE203	Course name	:	WEB DEVELOPMENT FRAMEWORKS

Practical – 9

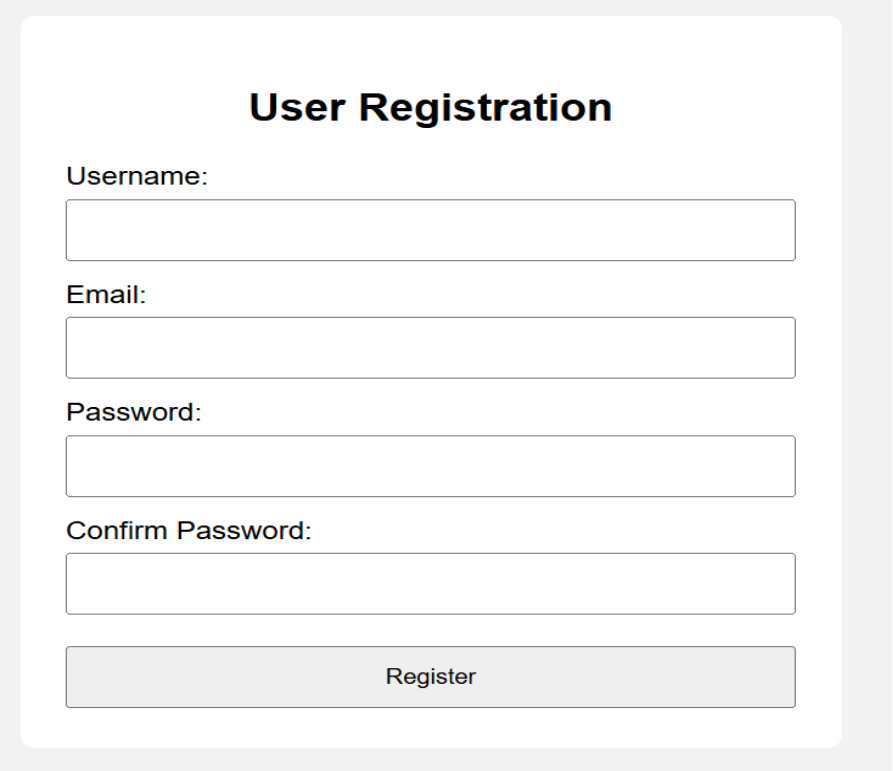
Secure User Registration with Database Insert, Duplicate Email Check, and Password Hashing

Problem Statement: - Implement secure user registration by inserting user details into MySQL using MYSQLi. Validate input on frontend and backend, check duplicate email/username, hash passwords using password_hash(), and display appropriate messages.

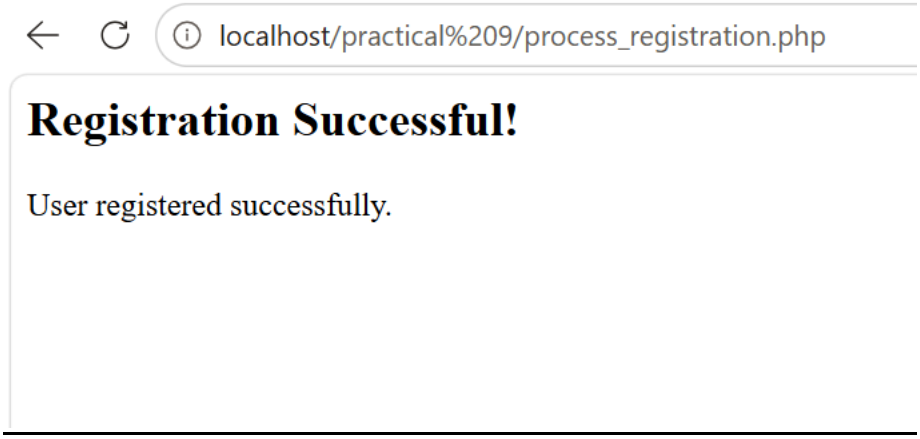
Objective :- To develop a secure user registration system using PHP and MySQL that validates user input, checks for duplicate email addresses and usernames, stores user details in the database using prepared statements, and protects passwords by hashing them using `password_hash()` before storage.

Screenshot

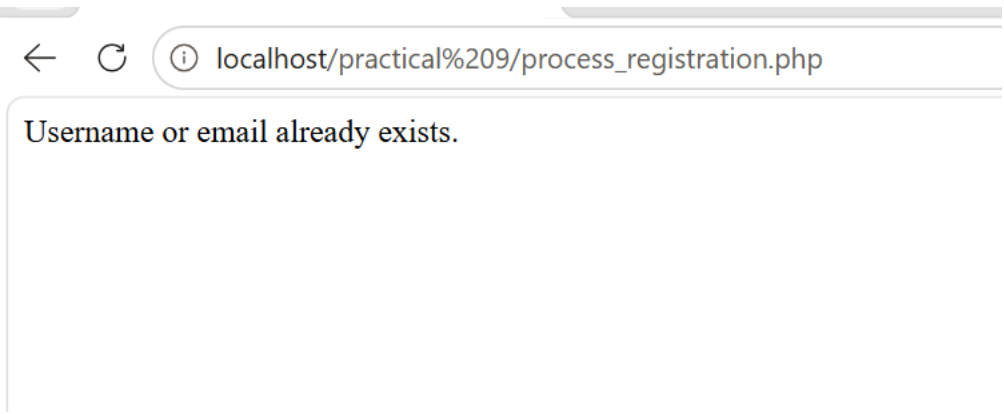
Registration form:



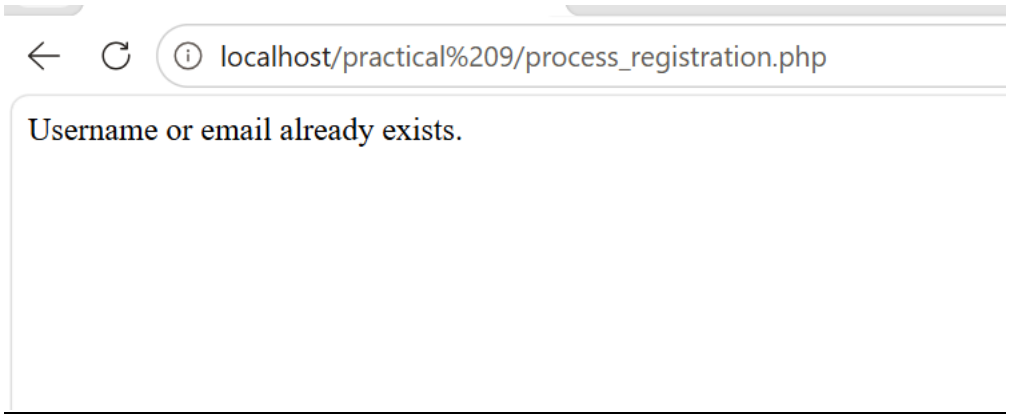
Successful registration:



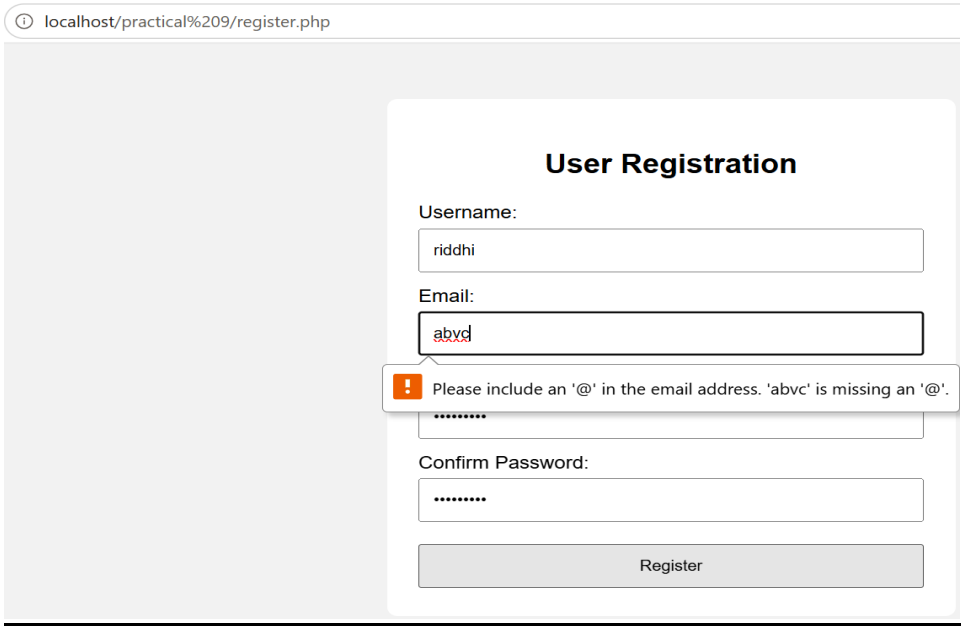
Duplicate email error :



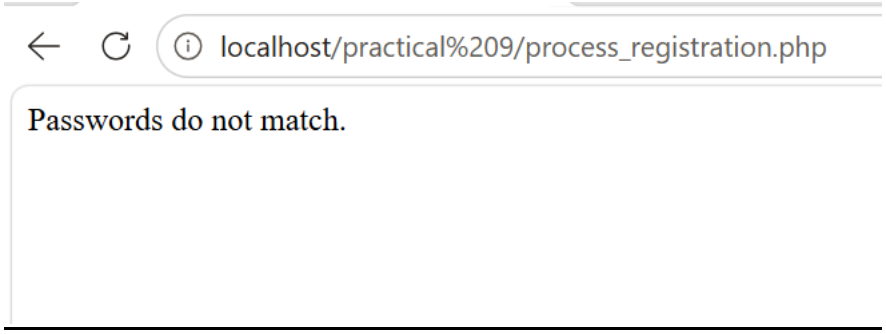
Duplicate username error :



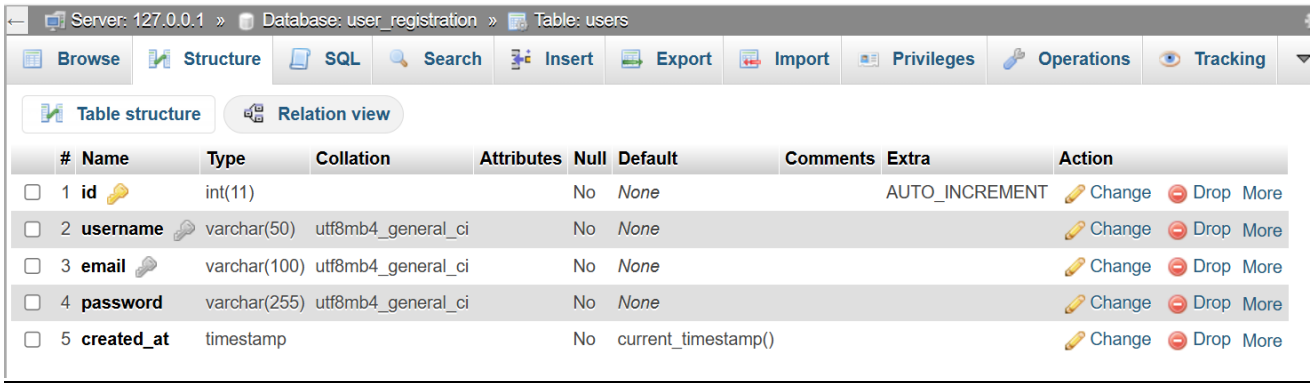
Invalid email validation:



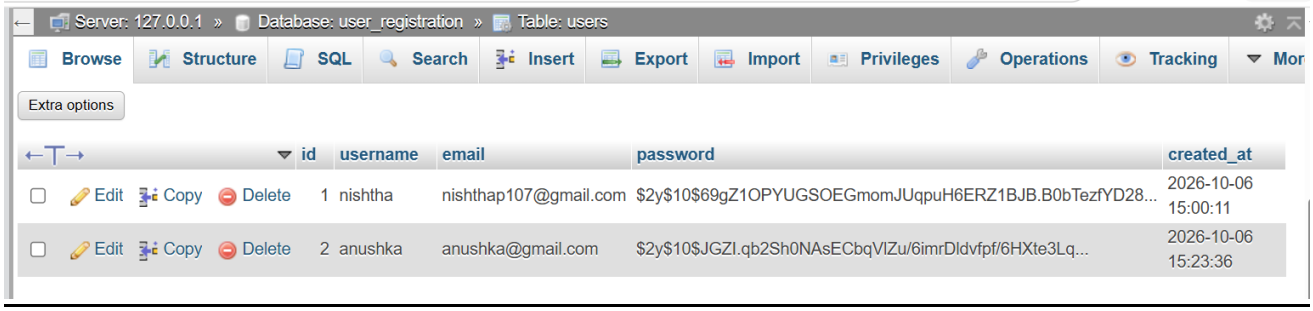
Password mismatch :



MySQL users table:



Hashed password visible in database:



Code:

db.php:

```
<?php

$host = "localhost";
$username = "root";
$password = "";
$database = "user_registration";

$conn = new mysqli($host, $username, $password, $database);

if ($conn->connect_error) {
    die("Database connection failed: " . $conn->connect_error);
}

$conn->set_charset("utf8mb4");

?>
```

Register.php:

```
<!DOCTYPE html>
<html>
<head>
    <title>User Registration</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

<div class="container">

    <h2>User Registration</h2>

    <form action="process_registration.php" method="POST">

        <label>Username:</label>
        <input
            type="text"
            name="username"
            required
            minlength="3"
            maxlength="50"
        >

        <label>Email:</label>
        <input
            type="email"
            name="email"
            required
        >

        <label>Password:</label>
        <input
            type="password"
            name="password"
            required
            minlength="6"
        >

        <label>Confirm Password:</label>
        <input
            type="password"
            name="confirm_password"
            required
            minlength="6"
        >

        <button type="submit">Register</button>

    </form>

</div>

</body>
</html>
```

Style.css:

```
body {
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
}

.container {
    width: 400px;
    margin: 50px auto;
    background: white;
    padding: 25px;
    border-radius: 8px;
}

h2 {
    text-align: center;
}

label {
    display: block;
    margin-top: 12px;
}

input {
    width: 100%;
    padding: 10px;
    margin-top: 5px;
    box-sizing: border-box;
}

button {
    width: 100%;
    padding: 10px;
    margin-top: 20px;
    cursor: pointer;
}
```

process_registration.php:

```
<?php

require_once "db.php";

if ($_SERVER["REQUEST_METHOD"] !== "POST") {
    die("Invalid request.");
}

// Get form data
$username = trim($_POST["username"] ?? "");
$email = trim($_POST["email"] ?? "");
$password = $_POST["password"] ?? "";
$confirm_password = $_POST["confirm_password"] ?? "";

// Backend validation
if ($username === "" || $email === "" || $password === "" || $confirm_password
=== "") {
    die("All fields are required.");
}

// Validate username
if (!preg_match("/^[a-zA-Z0-9_]{3,50}$/", $username)) {
    die("Username must contain only letters, numbers, and underscore.");
}
```

```
// Validate email
if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    die("Please enter a valid email address.");
}

// Validate password length
if (strlen($password) < 6) {
    die("Password must contain at least 6 characters.");
}

// Confirm password
if ($password !== $confirm_password) {
    die("Passwords do not match.");
}

// Check duplicate username/email
$check = $conn->prepare(
    "SELECT id FROM users WHERE username = ? OR email = ?"
);

$check->bind_param("ss", $username, $email);
$check->execute();
$check->store_result();

if ($check->num_rows > 0) {
    $check->close();
    $conn->close();
    die("Username or email already exists.");
}

$check->close();

// Hash password
$hashed_password = password_hash($password, PASSWORD_DEFAULT);

// Insert user
$stmt = $conn->prepare(
    "INSERT INTO users (username, email, password) VALUES (?, ?, ?)"
);

$stmt->bind_param(
    "sss",
    $username,
    $email,
    $hashed_password
);

if ($stmt->execute()) {

    echo "<h2>Registration Successful!</h2>";
    echo "<p>User registered successfully.</p>";

} else {

    echo "<h2>Registration Failed</h2>";
    echo "<p>Unable to register user.</p>";
}

$stmt->close();
$conn->close();

?>
```

Learning outcome:

- Implement a secure user registration system using PHP and MySQLi.
- Perform frontend and backend validation of user input.
- Check for duplicate usernames and email addresses before registration.
- Use `password_hash()` to securely hash user passwords before storing them.
- Insert user details into a MySQL database using secure prepared statements.
- Display appropriate success and error messages based on the registration result.